

Notes: Bernanke and Kuttner (JF, 2005)
What Explains the Stock Market's Reaction to Federal Reserve Policy?

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1 Big picture

- **Research question.** How does the stock market react to Federal Reserve monetary policy, and what economic channel explains that reaction?
- **Main empirical challenge.** Stock prices are forward-looking. The market should not react much to policy moves that were already expected. Therefore, the key object is not the raw change in the Federal funds target, but the *unexpected* component of that change.
- **Main measurement idea.** The paper uses Federal funds futures to decompose target-rate changes into expected and surprise components. The surprise component is the change in futures-implied expectations around the policy event.
- **Main finding.** A hypothetical unanticipated 25-basis-point cut in the Federal funds target is associated with about a 1% increase in broad stock indexes. Expected rate changes have little effect.
- **Main decomposition result.** The stock-market response is not mainly due to a mechanical real-interest-rate discounting channel. Instead, it is driven mostly by changes in expected future excess returns, with some role for expected dividends.
- **Cross-sectional implication.** Industry portfolios react differently. High-tech and telecommunications respond more strongly, while energy and utilities respond weakly. Much of this cross-industry pattern is consistent with market beta differences.
- **Economic takeaway.** Monetary policy affects stock prices because policy surprises change investors' views about risk premia, future cash flows, and the future policy path. The direct real-rate effect is surprisingly small.

2 Incremental contribution of the paper

- **From policy actions to policy surprises.** Earlier work often used raw interest-rate changes or VAR-identified policy shocks. Bernanke and Kuttner use Federal funds futures to isolate the unanticipated component of policy actions, which is the component equity markets should price on announcement days.
- **Clean event-study evidence.** The paper studies the union of target-rate-change days and FOMC meeting days from 1989 to 2002, allowing both policy actions and policy inactions to be treated as potentially informative events.
- **Robustness to outliers and endogeneity.** The paper explicitly studies influential observations, employment-report days, subsample stability, asymmetries, and the distinction between level surprises and timing surprises.
- **Daily and monthly consistency.** The paper shows that the negative relation between funds-rate surprises and stock returns is visible in both daily event windows and monthly data.
- **Industry-level evidence.** It goes beyond the aggregate index and estimates the response of Fama–French industry portfolios, then compares the pattern to CAPM-implied responses.

- **Economic-channel decomposition.** The paper adapts Campbell–Ammer return-decomposition methods to ask whether policy affects prices through dividends, real rates, or expected excess returns.
- **Key conceptual contribution.** The paper separates the *size* of the equity-market response from the *reason* for the response. The answer is that surprise policy matters, but mainly through discount-rate/risk-premium expectations rather than through current real rates alone.

3 Data and measurement

3.1 Federal funds futures and policy surprises

The key monetary-policy variable is the unexpected change in the Federal funds target. For an event on day d of month m , the surprise component is measured from the change in the current-month Federal funds futures rate:

$$\Delta i^u = \frac{D}{D-d} \left(f_{m,d}^0 - f_{m,d-1}^0 \right),$$

where D is the number of days in the month and $f_{m,d}^0$ is the current-month futures-implied funds rate.

The expected component is then:

$$\Delta i^e = \Delta i - \Delta i^u.$$

Interpretation. The futures-market change identifies the part of the policy action that was not already priced. If the FOMC delivers exactly what the market expected, the surprise component is small even if the target-rate change is large.

3.2 Daily event-study sample

- **Event days.** The sample is the union of days when the target Federal funds rate changed and FOMC meeting days.
- **Sample period.** June 1989 through December 2002.
- **Excluded observation.** September 17, 2001 is excluded because it was the first trading day after the September 11 attacks, making it impossible to isolate monetary-policy news.
- **Main return variable.** The one-day CRSP value-weighted equity return.
- **Number of observations.** The baseline event sample has 131 observations after excluding September 17, 2001.

Interpretation. The daily design is intended to isolate immediate stock-market repricing around policy news. It is closest to a standard event study.

3.3 Monthly data and industry portfolios

The paper also uses monthly data to compare the daily evidence with a regular-frequency time-series approach. The monthly surprise is defined as the average target rate in month t minus the

futures-implied expected average rate at the end of the previous month:

$$\bar{\Delta}i_t^u = \frac{1}{D} \sum_{d=1}^D i_{t,d} - f_{t-1,D}^1.$$

The expected monthly change is:

$$\bar{\Delta}i_t^e = f_{t-1,D}^1 - i_{t-1,D}.$$

The monthly analysis is also used for industry portfolios. The industry portfolios are the 10 Fama–French industry portfolios: nondurables, durables, manufacturing, energy, high-tech, telecommunications, wholesale/retail, health care, utilities, and other.

3.4 Economic-news controls

The paper studies whether funds-rate surprises are correlated with macroeconomic news, especially in the early part of the sample. It uses Money Market Services survey expectations to construct surprises in CPI, PPI, nonfarm payrolls, industrial production, and retail sales.

Interpretation. If weak employment news causes both a stock-market decline and a surprise Fed easing, a naive event-study regression can understate the positive effect of easing on stock prices.

3.5 VAR variables for the decomposition

To explain the stock-price reaction, the paper uses a monthly VAR with the following variables:

- excess equity return;
- real short rate;
- relative bill rate;
- change in the bill rate;
- smoothed dividend-price ratio;
- spread between the 10-year and 1-month Treasury yields.

Interpretation. The VAR is used to form expectations of future excess returns and real rates. Dividend news is backed out using the return identity.

4 Methodology and empirical specifications

4.1 Baseline daily event-study regression

The simplest specification regresses stock returns on the raw target-rate change:

$$H_t = a + b\Delta i_t + \varepsilon_t,$$

where H_t is the one-day stock return.

The main specification decomposes the target-rate change into expected and surprise components:

$$H_t = a + b^e \Delta i_t^e + b^u \Delta i_t^u + \varepsilon_t.$$

The coefficient b^u is the main event-study estimate. A negative value means that a surprise tightening lowers stock prices, or equivalently that a surprise easing raises stock prices.

4.2 Outliers and influence statistics

The paper computes influence statistics by dropping each observation in turn and measuring how much the estimated regression coefficients change. Six observations have unusually high influence:

- August 21, 1991;
- July 2, 1992;
- October 15, 1998;
- January 3, 2001;
- March 20, 2001;
- April 18, 2001.

Interpretation. These observations are informative because they reveal the sensitivity of the event-study coefficient to extreme market moves and unusual policy episodes. The paper reports results both with and without them.

4.3 Employment-report endogeneity

In the pre-1994 sample, several rate cuts occurred on employment-report days. The Fed may have been responding to bad employment news on the same day that equity prices also responded negatively to the news. The paper therefore estimates regressions with an interaction between the policy surprise and an employment-report dummy.

Interpretation. Controlling for employment-report days increases the estimated policy effect. This supports the view that same-day macro news biases the naive response toward zero.

4.4 Asymmetries

The paper tests whether the stock-market response differs by:

- positive versus negative policy surprises;
- actual rate increases versus cuts;
- policy inaction, meaning no target-rate change despite some market expectation of a change;
- FOMC meeting versus intermeeting actions;
- reversals in the direction of rate changes.

The strongest asymmetry is that the response is smaller, and often close to zero, for surprises associated with no target-rate change. The interpretation is that failure to move at a given meeting may mostly postpone an expected action rather than permanently change the expected rate path.

4.5 Timing versus level surprises

A policy surprise can change expected rates in two different ways. It can change only the *timing* of an anticipated move, or it can change the expected *level* of future short rates.

The paper measures this using the response of the three-month-ahead Federal funds futures rate. The timing-surprise variable is:

$$\text{Timing surprise} = \Delta f^{3\text{-month}} - \Delta i^u.$$

Interpretation. Stock prices respond much more to surprises that alter the expected level of future short rates. Pure timing surprises have little estimated effect.

4.6 Monthly stock-return regression

At monthly frequency, the paper estimates:

$$H_t = a + b^e \bar{\Delta} i_t^e + b^u \bar{\Delta} i_t^u + \varepsilon_t.$$

The monthly coefficient is larger in magnitude than the daily coefficient because the monthly surprise is measured as an average over the month. This time aggregation mechanically attenuates the size of the measured surprise, making the coefficient appear larger.

4.7 Industry-portfolio regressions

The same monthly regression is estimated separately for the 10 Fama–French industry portfolios. The paper then compares the estimated industry response to the response implied by the CAPM:

$$\hat{b}_i^u = \hat{\beta}_i \times \hat{b}^u,$$

where $\hat{\beta}_i$ is the industry’s market beta and $\hat{b}^u$ is the aggregate market response.

Interpretation. Industries with higher market betas should have larger absolute responses to policy surprises. The observed pattern is broadly consistent with this implication.

4.8 Campbell–Ammer return decomposition

The paper decomposes unexpected excess equity returns into news about dividends, real interest rates, and future excess returns:

$$e_{t+1}^y = \tilde{e}_{t+1}^d - \tilde{e}_{t+1}^r - \tilde{e}_{t+1}^y.$$

Here:

- $\tilde{e}_{t+1}^d$ is news about discounted future dividend growth;
- $\tilde{e}_{t+1}^r$ is news about current and future real rates;
- $\tilde{e}_{t+1}^y$ is news about discounted future excess returns.

The VAR is written as:

$$z_{t+1} = Az_t + w_{t+1}.$$

To study monetary policy, the paper adds the monthly funds-rate surprise:

$$z_{t+1} = Az_t + \phi \bar{\Delta} i_{t+1}^u + w_{t+1}^\perp.$$

Interpretation. The coefficient vector ϕ gives the contemporaneous response of the VAR variables to a policy surprise. The VAR dynamics then map this surprise into revisions in expected dividends, real rates, and future excess returns.

5 Identification and interpretation

5.1 Why expected and unexpected policy must be separated

If the market has already anticipated a rate cut, stock prices should have adjusted before the event day. Using the raw target-rate change mixes anticipated and unanticipated components and therefore understates the market reaction to genuine news.

5.2 Event-study orthogonality condition

The event-study regression assumes that other news affecting stock prices on event days is orthogonal to the measured policy surprise. This assumption can fail if monetary policy and the stock market respond simultaneously to the same macro news.

The paper argues that this problem, if anything, tends to attenuate the estimated response. For example, bad macro news may cause both a Fed easing and a stock-market decline, making easing look less positive for equities than it really is.

5.3 Employment-report days

The early sample contains several rate cuts on employment-report days. These are the clearest cases in which policy surprises may be endogenous to macro news. The employment-interaction regressions show that the stock-market response is much smaller on these days, consistent with bad employment news offsetting the positive valuation effect of surprise easing.

5.4 Outlier interpretation

The outliers are not simply data mistakes. Some are associated with unusual market conditions, such as the October 1998 intermeeting cut during financial-market stress and the large 2001 easing-cycle reactions. Reporting estimates with and without them is important because they show the range of plausible average effects.

5.5 Timing versus level interpretation

The paper's timing-versus-level analysis clarifies that equity markets care about the expected future path of policy, not just the current target. If a surprise cut merely brings forward a cut that investors already expected soon, the effect on equity values is small. If it lowers the expected path of rates for months ahead, the effect is larger.

5.6 What the VAR decomposition can identify

The Campbell–Ammer decomposition is an accounting identity combined with a VAR forecasting model. It tells us how the VAR attributes stock-return news to dividends, real rates, and future excess returns. It does not by itself prove a structural asset-pricing model.

Interpretation. The finding that expected excess returns explain much of the reaction can be read as a change in the equity premium, but it could also reflect market overreaction. The paper is careful about this distinction.

5.7 Interpretation of the expected-excess-return channel

A surprise tightening lowers current stock prices and is followed, in the VAR, by higher expected future excess returns. This is consistent with tight money raising required returns or the equity premium. Possible mechanisms include higher firm risk, weaker balance sheets, lower investor risk tolerance, or changes in expected consumption risk.

6 Tables and figures

6.1 Table I and Table II: descriptive statistics and baseline stock-market response

Read Table I:

- Table I reports descriptive statistics for policy surprises and equity returns before and after 1994.
- The standard deviation of Federal funds surprises is roughly 10 basis points in both periods.
- Equity returns are more volatile after 1994 than before 1994.
- Equity returns are slightly more volatile on policy event days than on non-event days.

Read Table II:

- The raw target-rate change has a small and insignificant relation with stock returns.
- Once the target-rate change is split into expected and surprise components, the surprise component is strongly significant.
- In the full sample, a 1 percentage point surprise tightening is associated with a -4.68% one-day return.

- Excluding influential outliers reduces the estimate to -2.55% , but it remains significant.
- The expected component has much weaker and less stable explanatory power.

Economic message: Equity prices react to unexpected monetary-policy news, not to the mechanical target-rate change itself. A 25-basis-point surprise easing implies roughly a 1% increase in broad stock prices in the full-sample estimate.

Table I
Descriptive Statistics

The table reports selected descriptive statistics for Federal funds rate surprises and the CRSP value-weighted equity return over the samples given in the column headings. All statistics exclude the September 17, 2001 observation.

	May 1989–January 1994	February 1994–December 2002
Number of events: rate changes and FOMC meetings	55	76
Standard deviation of Federal funds surprise, basis points	10.4	9.5
Standard deviation of equity return on event days, %	0.80	1.26
Standard deviation of equity return on nonevent days, %	0.71	1.11
Proportion of rate changes taking place at FOMC meetings	0.67	0.95

Table I: Descriptive statistics for funds-rate surprises and equity returns.

Table II
The Response of Equity Prices to Federal Funds Rate Changes

The table reports the results from regressions of the 1-day CRSP value-weighted equity return on changes in the Federal funds rate, columns (a) and (c), and on the surprise and expected components of the funds rate change, columns (b) and (d). All variables are expressed in percentage terms. The full sample consists of the 55 target rate changes and the 77 FOMC meeting dates over the period from June 1989 through December 2002, excluding the September 17, 2001 observation, for a total of 131 observations. The outliers excluded from the regressions in columns (c) and (d) correspond to the six observations with influence statistics in excess of 0.3, leaving 125 usable observations. Parentheses contain t -statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	Full Sample		Excluding Outliers	
	(a)	(b)	(c)	(d)
Intercept	0.23 (2.58)	0.12 (1.35)	0.17 (2.14)	0.11 (1.37)
Raw funds rate change	-0.61 (1.06)	–	-0.11 (0.31)	–
Expected change	–	1.04 (2.17)	–	0.67 (1.62)
Surprise change	–	-4.68 (3.03)	–	-2.55 (2.79)
R^2	0.007	0.171	-0.007	0.049

Table II: Baseline response of equity prices to Federal funds rate changes.

6.2 Figure 1 and Figure 2: daily event-study scatter and influential observations

Read Figure 1:

- Figure 1 plots one-day CRSP value-weighted returns against the Federal funds rate surprise.
- The scatterplot shows a negative relation: surprise tightening is associated with lower equity returns.
- Several events have large influence, especially in 1998 and 2001.

Read Figure 2:

- Figure 2 shows the distribution of influence statistics from dropping each observation in turn.
- Most observations have very low influence.
- Six observations stand out with influence statistics above 0.3.

Economic message: The daily event-study result is visible in the raw data, but a small number of extreme events materially affect the estimated magnitude. The qualitative conclusion survives their removal.

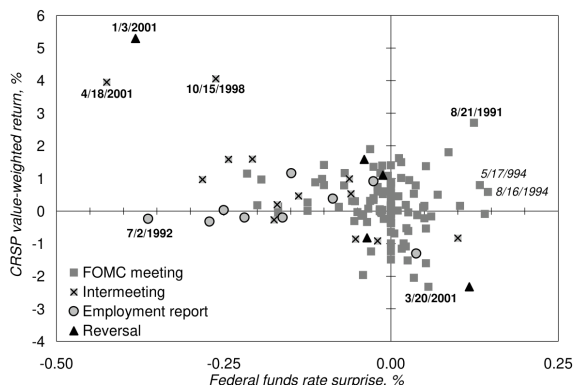


Figure 1: Federal funds rate surprises and one-day equity returns.

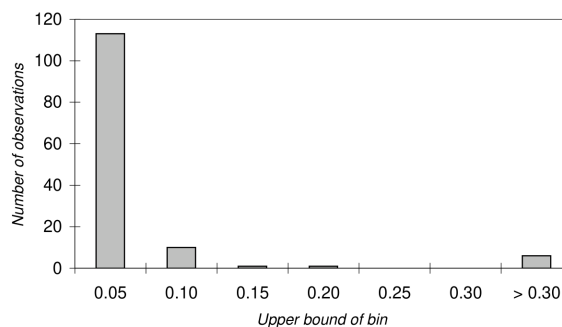


Figure 2: Distribution of regression influence statistics.

6.3 Table III and Table IV: subsample stability, endogeneity, and asymmetries

Read Table III:

- Table III tests whether the response differs after 1994 and on employment-report days.
- In the full sample, the post-1994 interaction is large, but this is influenced by outliers.
- Excluding outliers and controlling for employment-report days produces a surprise coefficient near -3.57 .
- The employment interaction is positive and nearly offsets the main surprise effect, consistent with bad employment news offsetting the valuation effect of easing.

Read Table IV:

- Table IV tests several forms of asymmetry.
- The response is not clearly different for positive versus negative surprises.
- Policy surprises associated with no target-rate change have a much smaller effect.
- Reversals sometimes have large estimated effects, but there are too few reversal observations for strong inference.

Economic message: The stock-market response is stable once outliers and same-day employment news are handled. Policy inaction is less powerful than action because it may change timing rather than the expected level of future policy.

Table III
Tests for Subsample Stability and Endogeneity

The table reports the results from regressions of the 1-day CRSP value-weighted equity return on the surprise and expected components of the change in the Federal funds rate, all expressed in percentage terms. The post-1994 dummy is set to 1 for observations beginning with February 4, 1994. The employment dummy is set to 1 for pre-1994 observations when a change in the target funds rate coincided with an employment release. The full and no-outlier samples are the same as those used for the results appearing in Table II. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	Full Sample		Excluding Outliers	
	(a)	(b)	(c)	(d)
Intercept	0.16 (1.80)	0.16 (1.76)	0.12 (1.43)	0.12 (1.43)
Expected change	1.09 (2.26)	1.09 (2.24)	0.69 (1.68)	0.69 (1.67)
Surprise change	-1.25 (1.14)	-2.55 (1.70)	-2.29 (2.28)	-3.57 (3.77)
Surprise change × post-1994	-6.87 (3.59)	-5.58 (2.61)	-0.78 (0.45)	0.50 (0.29)
employment report	-	2.67 (1.82)	-	3.33 (2.55)
R^2	0.280	0.283	0.042	0.054

Table III: Subsample stability and employment-report endogeneity.

Table IV
Tests for Asymmetries

The table reports the results from regressions of the 1-day CRSP value-weighted equity return on the surprise and expected components of the change in the Federal funds rate, all expressed in percentage terms. The positive surprise dummy is set to 1 when the surprise change in the funds rate is greater than 0. The no rate change and positive rate change dummies equal 1 when the funds rate target is unchanged or increased. The FOMC meeting dummy is set to 1 for those observations coinciding with FOMC meetings. The reversal dummy equals 1 for rate changes that reverse the direction of the previous change. The post-1994 dummy is set to 1 for observations beginning with February 4, 1994. The employment dummy is set to 1 for pre-1994 observations when a change in the target funds rate coincided with an employment release. The full and no-outlier samples are the same as those used for the results appearing in Table II. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	Full Sample			Excluding Outliers		
	(a)	(b)	(c)	(d)	(e)	(f)
Intercept	-0.02 (0.17)	0.12 (1.34)	0.14 (1.72)	0.12 (1.32)	0.12 (1.42)	0.13 (1.63)
Expected change	0.84 (1.58)	1.56 (3.24)	1.03 (2.24)	0.72 (1.67)	0.97 (2.00)	0.72 (1.76)
Surprise change	-7.57 (4.67)	-8.34 (5.73)	-3.97 (2.98)	-3.26 (3.30)	-4.49 (4.91)	-3.67 (3.14)
Surprise change × employment	7.05 (4.43)	8.11 (5.26)	2.54 (1.47)	3.05 (2.36)	4.14 (3.19)	3.46 (2.35)
positive surprise	7.39 (1.59)	-	-	-0.34 (0.10)	-	-
no rate change	-	10.42 (3.81)	-	-	4.00 (2.25)	-
positive rate change	-	3.05 (0.76)	-	-	0.58 (0.15)	-
FOMC meeting	-	-	4.25 (2.75)	-	-	0.67 (0.39)
reversal	-	-	-6.33 (3.09)	-	-	-17.62 (4.08)
post-1994	-	-	-4.61 (2.48)	-	-	0.80 (0.44)
R^2	0.260	0.323	0.369	0.053	0.065	0.098

Table IV: Tests for asymmetries in the event-study response.

6.4 Figure 3, Table V, and Table VI: timing versus level surprises

Read Figure 3:

- Figure 3 compares the current-month policy surprise with the change in the three-month-ahead Federal funds futures rate.
- Many surprises have less than a one-for-one effect on expected rates three months ahead.
- This indicates that many policy surprises are partly timing surprises.

Read Table V:

- A 1 percentage point current policy surprise changes the three-month-ahead futures rate by about 0.65 percentage points.
- The response is smaller for no-change surprises and FOMC meeting surprises.

Read Table VI:

- Adding the timing-surprise variable increases the estimated effect of a level surprise.
- A surprise that affects the expected level of rates has a larger stock-price effect than a pure timing surprise.
- Pure timing surprises have no statistically significant effect on equity prices.

Economic message: The stock market prices the future path of monetary policy. It reacts most when a policy surprise changes expected rates beyond the immediate meeting.

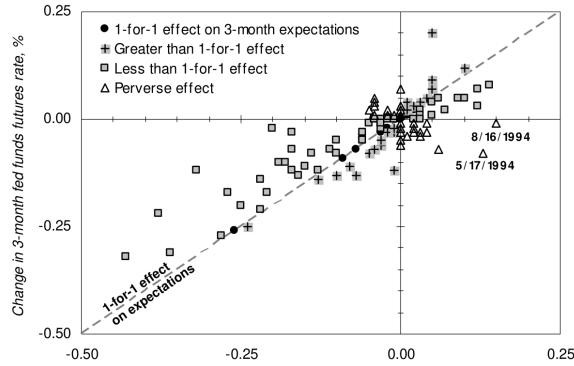


Figure 3: Current policy surprises and expected future funds rates.

Table VI
The Stock Market Response to Level versus Timing Surprises

The table reports the results from regressions of the 1-day CRSP value-weighted equity return on the surprise and expected components of the change in the Federal funds rate and the timing surprise, all expressed in percentage terms. The timing surprise is defined as the difference between the change in the 3-month-ahead futures rate and the current month's surprise. The full and no-outlier samples are the same as those used for the results appearing in Table II. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	Full Sample		Excluding Outliers	
	(a)	(b)	(c)	(d)
Intercept	0.12 (1.35)	0.09 (1.09)	0.11 (1.37)	0.09 (1.11)
Expected change	1.05 (2.17)	1.34 (2.92)	0.67 (1.62)	0.94 (2.46)
Surprise change	-4.68 (3.03)	-6.20 (3.80)	-2.55 (2.79)	-4.17 (4.20)
Timing surprise	—	-4.29 (2.20)	—	-4.27 (3.25)
Effect of "pure" timing surprise	—	-1.91 (0.91)	—	0.09 (0.08)
R^2	0.171	0.192	0.049	0.085

Table VI: Stock-market response to level versus timing surprises.

Table V
The Response of Interest Rate Expectations to Federal Funds Rate Surprises

The table reports the results from regressions of the 1-day change in the 3-month-ahead Federal funds futures rate on the surprise and expected components of the change in the Federal funds rate, all expressed in percentage terms. The no rate change and positive rate change dummies equal 1 when the funds rate target is unchanged or increased. The FOMC meeting dummy is set to 1 for those observations coinciding with FOMC meetings. The reversal dummy equals 1 for rate changes that reverse the direction of the previous change. The full and no-outlier samples are the same as those used for the results appearing in Table II. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	(a)	(b)	(c)	(d)
Intercept	-0.01 (1.46)	-0.01 (1.55)	-0.01 (1.34)	-0.01 (1.40)
Expected change	0.07 (2.10)	0.05 (1.32)	0.07 (2.29)	0.07 (2.08)
Surprise change	0.65 (13.37)	0.70 (14.71)	0.73 (14.54)	0.66 (12.83)
Surprise change $\times$ no rate change	—	-0.36 (3.24)	—	—
FOMC meeting	—	—	-0.21 (2.07)	—
reversal	—	—	—	-0.12 (2.24)
R^2	0.726	0.745	0.744	0.727

Table V: Response of three-month-ahead rate expectations to policy surprises.

Table VII
The Impact of Economic News on Federal Funds Rate Surprises

The table reports the results from regressions of the monthly Federal funds rate surprise on the unexpected components of the data releases listed in the row headings, over the sample indicated in the column headings. Survey data gathered by Money Market Services are used to calculate the data surprises.

Data surprise	Full sample	Subsample	
		5/89–9/92	2/94–12/02
Headline CPI	0.016	-0.124	-0.010
Core CPI	-0.058	-0.012	0.152
Headline PPI	0.001	-0.027	-0.024
Core PPI	-0.085**	-0.304***	-0.022
Nonfarm payrolls	0.203**	0.624***	-0.009
Industrial production	0.069*	0.136	0.028
Retail sales	-0.031**	-0.061	-0.035***
Retail sales, x autos	0.023	0.093	0.021
R^2	0.128	0.454	0.087
$\bar{R}^2$	0.082	0.304	0.012

Asterisks denote statistical significance based on heteroskedasticity-consistent estimates of the standard errors: *** for the 0.01 level, ** for the 0.05 level, and * for the 0.01 level.

Table VII: Economic-news predictors of monthly funds-rate surprises.

6.5 Table VII, Table VIII, and Figure 4: monthly policy surprises

Read Table VII:

- Table VII shows that monthly policy surprises are related to macroeconomic news, especially in the early 1989–1992 period.
- Nonfarm payroll surprises strongly predict funds-rate surprises in the early sample.
- The relationship is much weaker after 1994, when FOMC meetings made the timing of policy actions more regular.

Read Table VIII:

- Monthly stock returns also respond negatively to unanticipated funds-rate changes.
- The full-sample monthly coefficient is -11.43 .

- This coefficient is larger than the daily coefficient because monthly average surprises are attenuated by time aggregation.
- Dropping monthly outliers does not weaken the result.

Read Figure 4:

- Figure 4 shows the negative monthly relation between policy surprises and stock returns.
- The scatterplot includes candidate outliers, but the relationship remains visible.

Economic message: The daily event-study result is not an artifact of the particular event-day sample. A similar relation appears at monthly frequency, although monthly surprises raise additional endogeneity and time-aggregation issues.

Table VIII
The Monthly Response of Equity Prices to Federal Funds Rate Surprises

The table reports the results from regressions of the 1-month CRSP value-weighted equity return on the surprise and expected components of the 1-month change in the Federal funds rate, all expressed in percentage terms. The full sample includes 164 monthly observations spanning May 1989 through December 2002. The no-outlier sample contains 154 observations. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Regressor	Full Sample (a)	No Outliers (b)	Tests for Asymmetries		
			(c)	(d)	(e)
Intercept	0.13 (0.32)	-0.03 (0.09)	-0.01 (0.02)	-0.07 (0.16)	0.10 (0.24)
Expected change	-1.11 (0.37)	0.96 (0.35)	-1.07 (0.36)	-2.72 (0.72)	-1.09 (0.36)
Surprise change	-11.43 (3.95)	-14.26 (5.43)	-12.46 (3.69)	-11.01 (3.46)	-10.49 (2.53)
Surprise change × positive surprise	—	—	6.82 (0.63)	—	—
no rate change	—	—	—	-4.88 (0.75)	—
positive rate change	—	—	—	6.59 (0.52)	—
reversal	—	—	—	—	3.52 (0.50)
post-1994	—	—	—	—	-3.77 (0.50)
Employment surprise	—	—	—	—	-0.69 (0.10)
R^2	0.065	0.096	0.061	0.056	0.049
Standard error	4.28	3.85	4.30	4.30	4.31
Durbin-Watson statistic	2.02	2.09	2.02	2.02	2.03

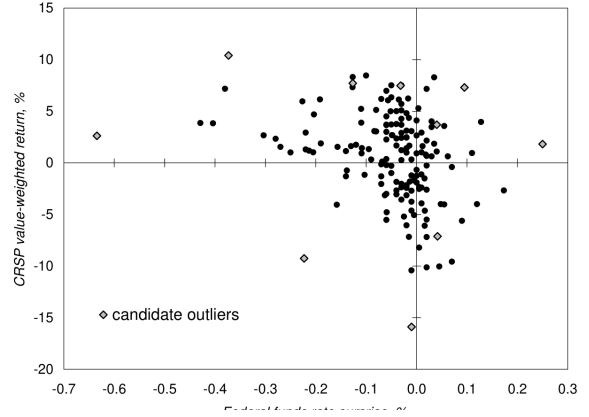


Table VIII: Monthly response of equity prices to Federal funds rate surprises.

Figure 4: Monthly funds-rate surprises and equity returns.

6.6 Table IX and Figure 5: industry portfolios and CAPM implications

Read Table IX:

- Table IX estimates the monthly response for 10 Fama–French industry portfolios.
- High-tech and telecommunications have the largest responses to unanticipated funds-rate changes.
- Energy and utilities have much smaller and statistically insignificant responses.
- The market betas line up with much of the cross-industry pattern.

Read Figure 5:

- Figure 5 plots estimated industry responses against CAPM-implied responses.
- The points broadly follow the 45-degree line.

- This suggests that one-factor market beta explains much of the heterogeneity in policy sensitivity across industries.

Economic message: Monetary policy is not equally important for all sectors. Sectors with higher market risk tend to have stronger policy sensitivity, consistent with standard asset-pricing logic.

Table IX
The Response of Fama–French Industry Portfolios to Federal Funds Rate Surprises

The table reports the results from regressions of the 1-month returns on the Fama–French industry portfolios indicated in the row headings on the surprise and expected components of the 1-month change in the Federal funds rate, all expressed in percentage terms. The regressions also include an intercept, whose coefficient is not reported. The full sample includes 164 monthly observations spanning May 1989–December 2002. Parentheses contain *t*-statistics, calculated using heteroskedasticity-consistent estimates of the standard errors.

Index	Response to Federal Funds Rate Changes		R^2	SE	DW	Market Beta
	Anticipated	Unanticipated				
CRSP value weighted	-1.11 (0.37)	-11.43 (3.95)	0.065	4.28	2.02	1
Nondurables	-0.85 (0.25)	-9.65 (2.88)	0.046	4.17	2.00	0.60
Durables	-1.47 (0.38)	-12.45 (3.04)	0.048	5.56	1.97	1.02
Manufacturing	-2.02 (0.61)	-8.82 (2.81)	0.035	4.26	2.03	0.85
Energy	0.20 (1.02)	-4.03 (1.24)	-0.003	4.71	2.12	0.55
High-tech	0.06 (0.01)	-14.73 (2.72)	0.025	8.22	2.00	1.61
Telecommunications	0.35 (0.60)	-16.10 (3.31)	0.065	6.16	1.85	1.16
Wholesale/retail	-4.75 (1.47)	-11.97 (3.64)	0.056	4.85	1.95	0.90
Health care	-1.04 (0.29)	-8.04 (1.80)	0.017	4.96	2.15	0.72
Utilities	-1.24 (0.48)	-5.42 (1.55)	0.006	4.21	1.97	0.32
Other	-1.21 (0.35)	-11.08 (3.61)	0.051	4.62	2.09	0.92

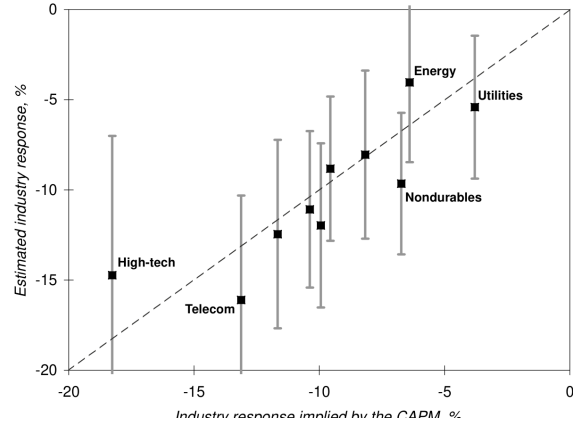


Table IX: Industry-portfolio responses to Federal funds rate surprises.

Figure 5: Estimated industry responses and CAPM-implied responses.

6.7 Table X, Figure 6, and Table XI: decomposition of the policy effect

Read Table X:

- Table X decomposes excess equity-return variance into dividend news, real-rate news, future-excess-return news, and covariance terms.
- Over 1973–2002, future excess returns account for the majority of the excess-return variance.
- The direct real-rate component is very small.
- In the 1989–2002 subsample, dividends play a larger role because equity-return forecastability is lower.

Read Figure 6:

- Figure 6 shows the dynamic response of the VAR variables to a Federal funds rate surprise.
- A surprise tightening produces a large immediate negative excess equity return.
- Real short-rate effects are sharp but transitory.
- Expected future excess returns become positive and persistent after the initial stock-price decline.

Read Table XI:

- Table XI decomposes the impact of a monetary-policy surprise on current excess returns.
- For the 1973–2002 VAR sample, the current excess-return response is about -11.55 .
- Future excess returns account for about 6.10 percentage points of this response.
- Dividends account for about -4.82 percentage points.
- The real-rate component is only about 0.64 percentage points.

Economic message: The direct real-rate discounting channel is too small to explain the stock-market response. Policy surprises primarily move stock prices by changing expected future excess returns and, to a lesser extent, expected dividends.

Table X
A Variance Decomposition of Excess Equity Returns

The table reports the decomposition of the variance of the current excess equity returns into the variances of revisions in expectations of dividends, real interest rates, future excess returns, and the covariances between these three components. The excess equity return is the difference between the CRSP value-weighted return and the 1-month Treasury bill rate. A six-variable VAR(1) is used to construct forecasts of future real interest rates and excess returns. The VAR includes the excess equity return, the real interest rate, the relative bill rate (defined as the 3-month bill rate minus its 12-month lagged moving average), the change in the 3-month bill rate, the smoothed dividend price ratio, and the spread between the 10-year and 1-month Treasury yields. Parentheses contain t -statistics, calculated using the delta method.

	1973–2002		1989–2002	
	Total	Share (%)	Total	Share (%)
Var(excess return)	21.5		19.0	
Var(dividends)	5.3	24.5 (6.2)	6.1	31.9 (1.8)
Var(real rate)	0.3	1.4 (2.4)	0.1	0.6 (1.5)
Var(future returns)	16.4	76.0 (1.8)	7.2	38.0 (1.2)
–2 Cov(dividends, real rate)	–0.4	–2.1 (0.8)	–0.6	–3.2 (0.7)
–2 Cov(dividends, future excess return)	0.2	1.0 (0.0)	7.2	–37.7 (2.3)
2 Cov(future excess return, real rate)	–0.2	0.8 (0.1)	1.0	5.1 (1.1)
R^2 from excess return equation	0.040		–0.003	

Table X: Variance decomposition of excess equity returns.

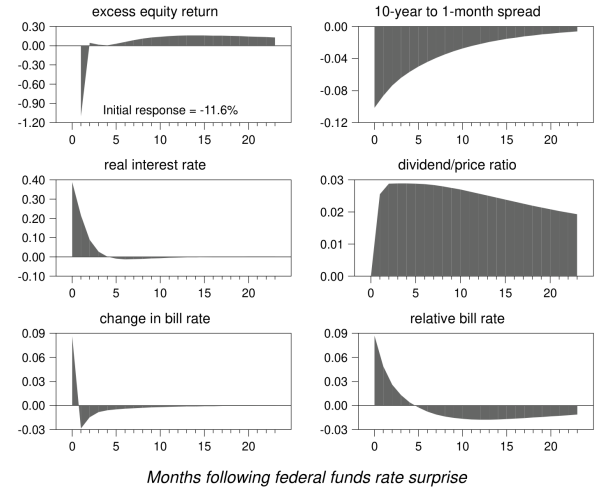


Figure 6: Dynamic VAR responses to a Federal funds rate surprise.

Table XI
The Impact of Monetary Policy on Dividends, Interest Rates, and Future Returns

The table reports the impact of monetary policy surprises on the current excess equity return and the discounted sums of future excess equity returns, current and future real interest rates, and current and future dividends. The six-variable VAR(1) used to construct real interest rate and excess equity return forecasts is estimated over the sample indicated in the column headings, and the contemporaneous response to the funds rate surprises is estimated on the February 1994–December 2002 subsample. Parentheses contain t -statistics, calculated using the delta method.

	Sample Used for VAR	
	1/73–12/02	5/89–12/02
Current excess return	–11.55 (3.87)	–11.01 (3.72)
Future excess returns	6.10 (1.74)	3.29 (1.10)
Real interest rate	0.64 (1.03)	0.77 (1.87)
Dividends	–4.82 (1.73)	–6.96 (2.35)

Table XI: Impact of monetary policy on dividends, real rates, and future excess returns.

7 Short summary

- The paper studies how the stock market responds to Federal Reserve policy actions.
- The key measurement step is separating expected policy changes from surprise changes using Federal funds futures.
- The market reacts strongly to surprise policy changes but weakly to expected changes.
- In the daily event-study sample, a surprise 25-basis-point easing raises broad stock indexes by about 1% in the full-sample estimate.
- The result is robust to excluding influential observations, although the estimated magnitude becomes smaller.
- Same-day employment news can bias estimates toward zero because weak employment news can trigger both Fed easing and lower stock prices.
- The market reacts more strongly to policy surprises that change the expected level of future rates, not merely the timing of an expected move.
- Monthly regressions confirm the negative relation between funds-rate surprises and equity returns.
- High-tech and telecommunications are especially responsive to policy surprises; energy and utilities are less responsive.
- The industry pattern is broadly consistent with CAPM market betas.
- A Campbell–Ammer VAR decomposition shows that the policy effect on stock prices is not mainly explained by current and future real rates.
- The largest channel is expected future excess returns, with a secondary role for expected dividends.
- The expected-excess-return result can be interpreted as a policy-induced change in the equity premium, but it may also reflect market overreaction.

8 Conclusion

8.1 Core conclusion

Bernanke and Kuttner show that equity markets react strongly to unexpected Federal Reserve policy actions. A surprise easing raises stock prices, while a surprise tightening lowers them. The reaction is economically large but depends crucially on whether the policy action was anticipated.

8.2 Economic intuition

Stock prices are present values of future payoffs discounted by required returns. A monetary-policy surprise can affect stock prices by changing expected dividends, real interest rates, or expected excess returns. The paper finds that expected excess returns are the dominant channel, while real-rate news is surprisingly small.

8.3 Incremental contribution revisited

The paper's main contribution is not simply showing that policy affects stocks. Its contribution is to isolate policy surprises with Federal funds futures and then decompose the valuation effect. This turns a broad policy-market correlation into evidence about the transmission channel from monetary policy to equity prices.

8.4 Implications

The findings imply that monetary policy works partly through asset prices and risk premia. Equity-market reactions to policy should not be interpreted as simple mechanical discount-rate effects. They also reflect changing views about future risk compensation and future cash flows.

8.5 Limitations

The event-study sample is modest in size and contains several influential observations. The futures-based surprise measure begins only in 1989. Same-day macroeconomic news can complicate identification, especially before 1994. The VAR decomposition relies on a forecasting model and cannot fully distinguish a rational equity-premium response from market overreaction.

8.6 Takeaway

The enduring takeaway is that the stock market responds to monetary policy because policy surprises change expectations. What matters most is not the target-rate change itself, but the news it conveys about the future path of policy, risk premia, and the economy.